

DANA ROCHA

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TECHNICAL SKILLS

Languages: Python, JavaScript, HTML, CSS, SQL, R

Frameworks & Technologies: React, Shiny, Flask, REST APIs, Apache Airflow

Databases: PostgreSQL, MongoDB, Google Firestore

Developer Tools: Git, GitHub, AWS, VS Code, Unix/Linux, REST APIs, CI/CD, Docker, RStudio, Firebase

WORK EXPERIENCE

Allen Institute

Seattle, WA

Data Analyst I - Allen Institute for Brain Science

Mar 2023 – Present

- Modernizing a legacy R-based scientific data pipeline processing 4.4+ TB of multimodal data by modularizing workflows, adding metadata validation, automated testing, and reducing manual processing by 10 hours per month
- Designed and prototyped an Apache Airflow-based data ingestion and orchestration pipeline to automate cloud-based omics workflow execution on AWS
- Optimized SQL queries against relational data sources to integrate research metadata, improve data traceability, and support downstream reporting and dataset delivery
- Developed a user-friendly Shiny web application, improving weekly data analysis by 30% through automated data updates and faster access to metadata
- Automated and streamlined data report generation and exports within an internal Python package
- Improved front-end functionality across 10+ Shiny web applications centralizing data extraction, visualization, and analysis, increasing usability and stakeholder adoption through iterative feedback and collaboration
- Established the team's first software engineering practices for internal tools, including CI/CD with GitHub Actions, automated testing, linting, formatting, and standardized development workflows
- Led and planned 30+ Women in Tech Employee Resource Group (ERG) programs for 2024-2026, managed a \$10k annual budget, coordinated stakeholders and resources to deliver high-impact company-wide initiatives

Software Engineer Intern - Allen Institute for Neural Dynamics

Aug 2022 – Jan 2023

- Led end-to-end development of a Minimum Viable Product (MVP) data-sharing web application, translating stakeholder requirements into wireframes and a production-ready prototype
- Built a responsive React front-end and implemented a service layer for REST API integration
- Established automated testing and CI with Jest, ESLint, and GitHub Actions
- Automated credential file generation to an open-source Python package, simplifying secure resource access
- Integrated third-party REST API dataset search into an open-source Python package, improving data discoverability
- Added JSON schema validation and export capabilities into an open-source Python package, enabling users to define reusable metadata templates

Merck & Co.

Boston, MA

Informatics Intern

Jan 2020 – Aug 2020

- Trained machine learning models to predict high-value compounds for early-phase testing, supporting data-driven decisions for three active drug discovery teams
- Built ETL pipelines using Oracle DBMS, SQL, Pipeline Pilot, PilotScript, and Spotfire to integrate data from public and internal sources, enabling downstream analysis and supporting data-driven decision making across the organization
- Presented monthly data insights to 50+ stakeholders, supporting alignment across drug-discovery teams

TECHNICAL PROJECTS

Room Reservation System, Capstone project – [GitHub](#)

Aug 2022

Web application to track room reservation information - developed with JavaScript, React, Google Firebase

- Collaborated closely in a two-person team to design, develop, and successfully deliver the project
- Wrote backend logic, database endpoints, and secure user authentication
- Crafted algorithms to create, delete, and prevent double-booking of reservations
- Established a responsive user interface of 3 pages utilizing Materialize and React Router for optimal user experience

EDUCATION

Ada Developers Academy

Seattle, WA

Full Stack Web Development & Computer Science Fundamentals

Mar 2022 – Jan 2023

Technologies: Python, JavaScript, SQL, PostgreSQL, Flask, React

Northeastern University

Boston, MA

B.S. in Biochemistry, M.S. in Bioinformatics

2018, 2020

Relevant Courses: Algorithms, Data Mining/Machine Learning, Python Programming